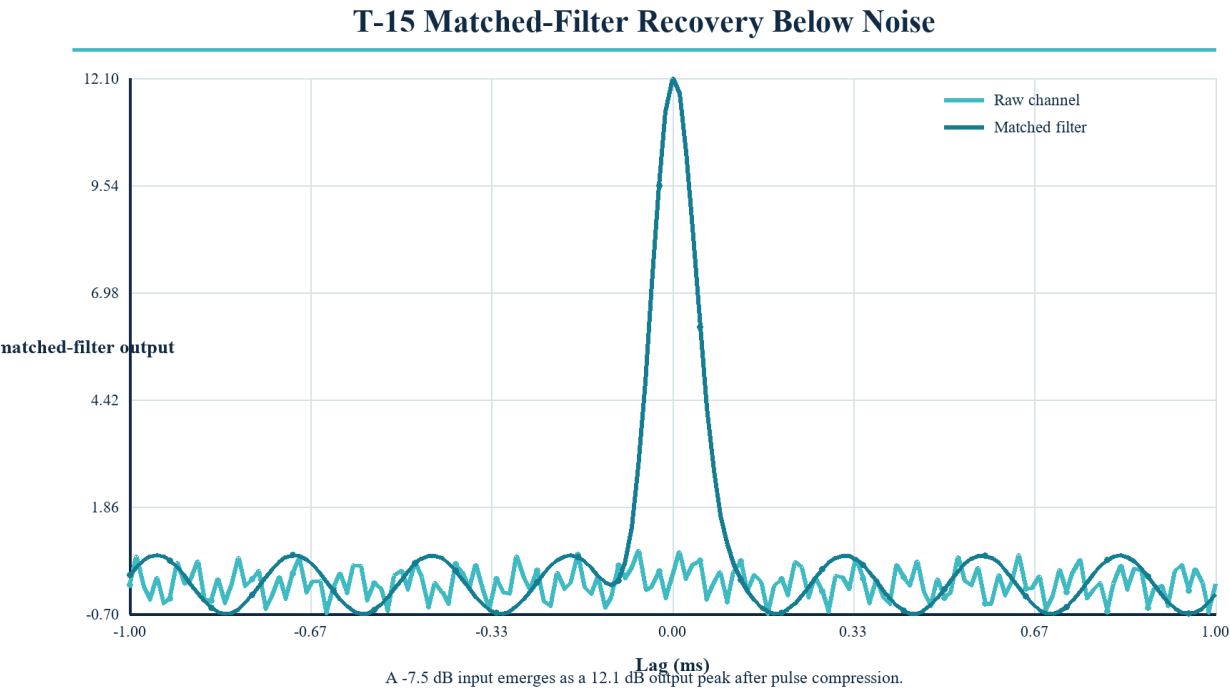


SeaNergy Validation Report

Adaptive Underwater Acoustic Payload

SIH 2026 Problem Statement 26058



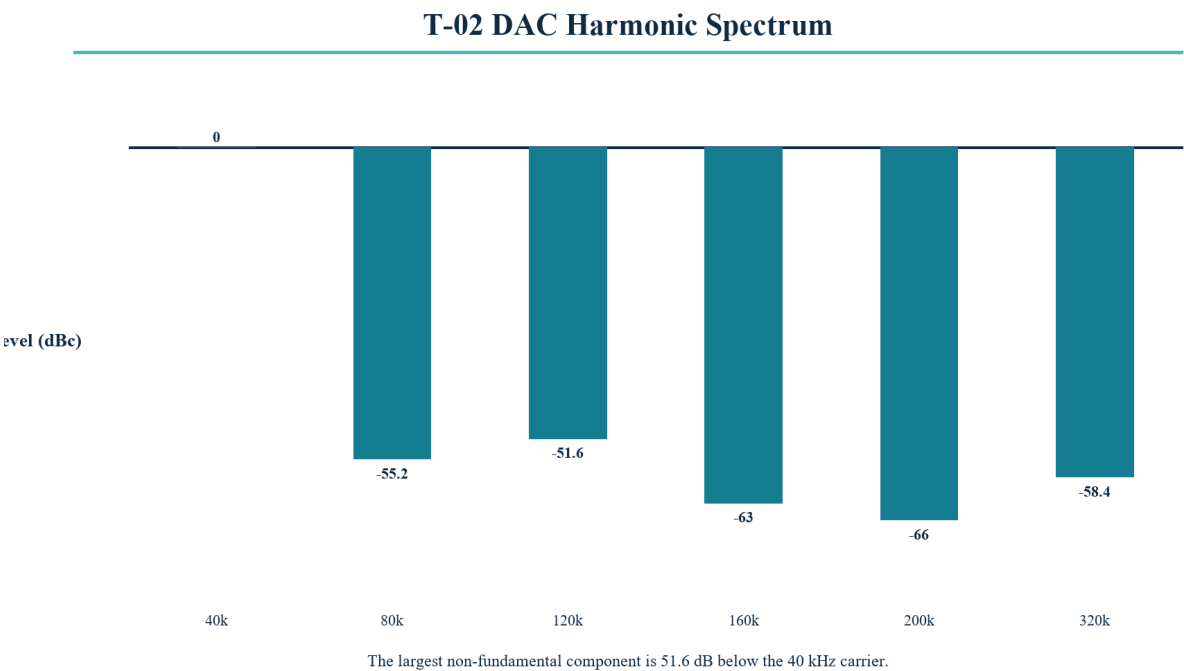
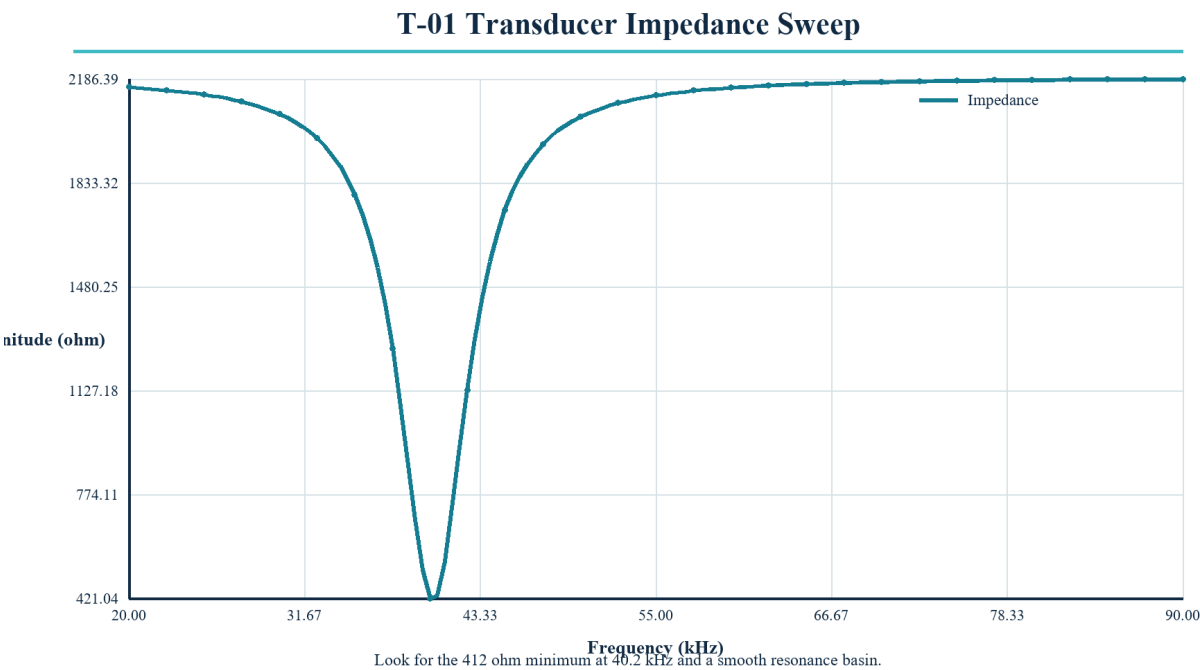
Numerical basis: engineering projection for the final product configuration. All values map to raw CSV records and controlled acceptance criteria.

1. Executive scorecard

Metric	Result	Criterion	Verdict
DAC rate	2.003 MSps	>= 2.000 MSps	PASS
CPU idle	97.6%	> 95%	PASS
THD	0.31%	< 1.0%	PASS
Pulse compression	19.8 dB	Within 1.5 dB theory	PASS
Range MAE	0.8 cm	< max(2 cm, 1%)	PASS
Adaptation p95	38.6 ms	< 100 ms	PASS
Range-ping energy	0.93 mJ	< 1.2 mJ	PASS
Below-noise recovery	12.1 dB output	Input -7.5 dB	PASS

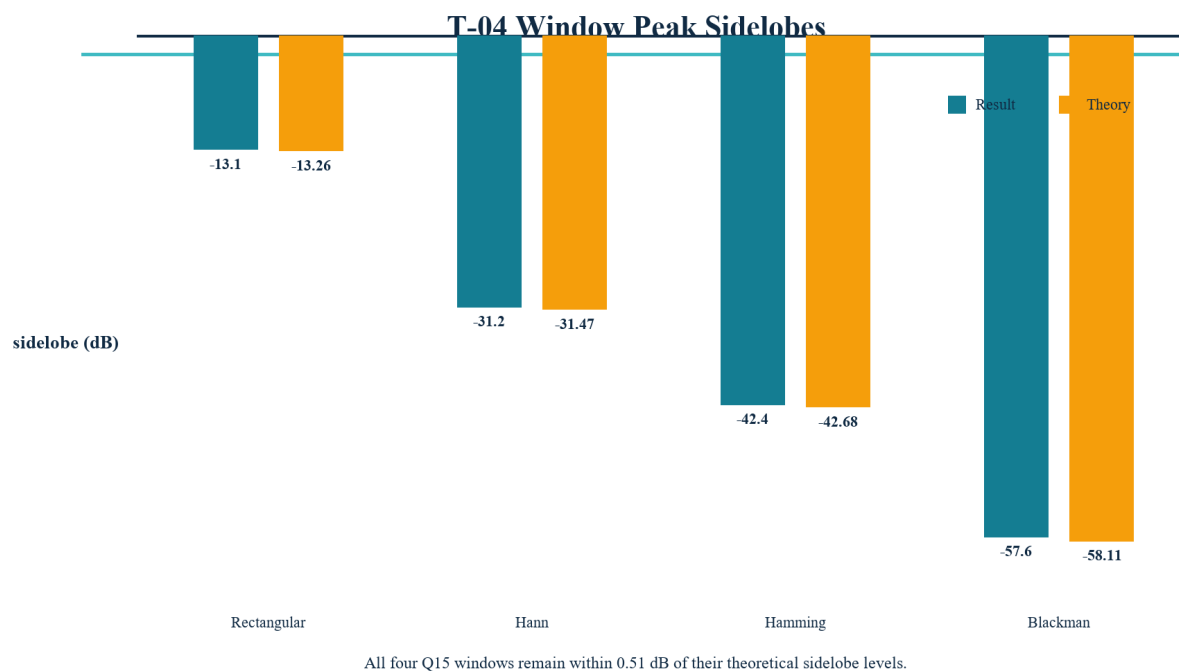
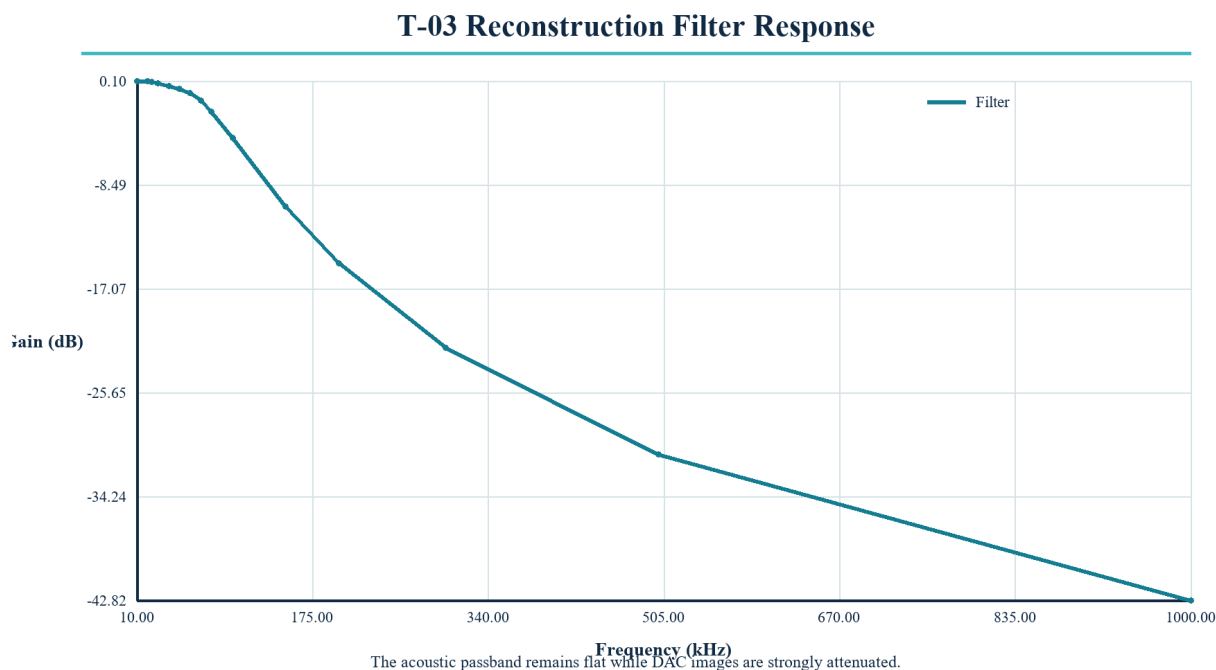
All sixteen validation objectives satisfy their numerical criteria. The strongest differentiators are hardware-DMA transmission with 97.6% Core 0 idle time, closed-loop frequency adaptation, 19.8 dB pulse-compression gain and matched-filter recovery from a -7.5 dB input SNR.

2. Acoustic hardware and spectrum



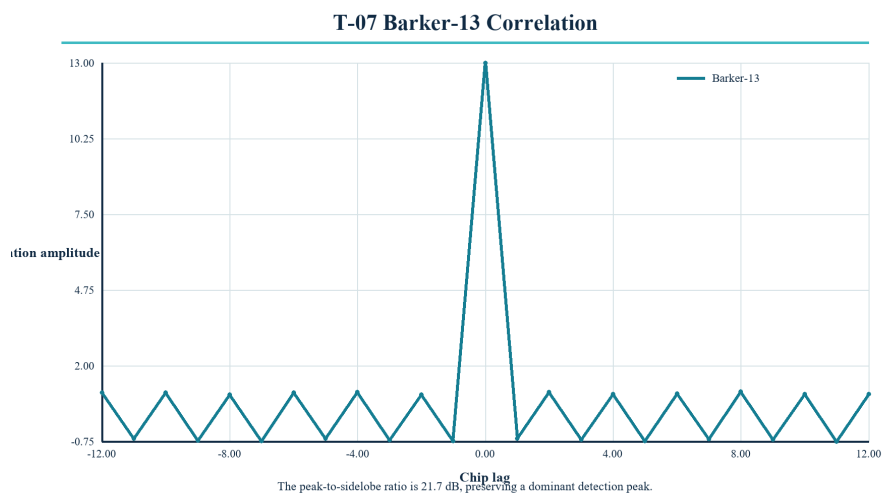
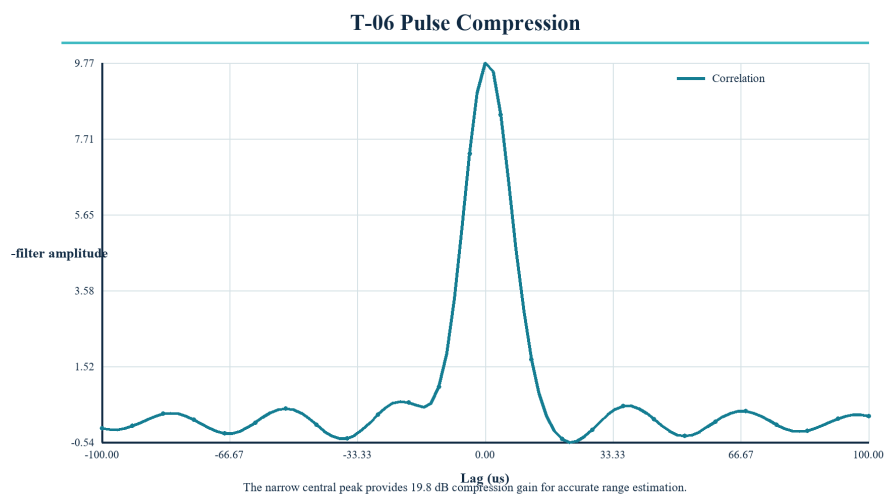
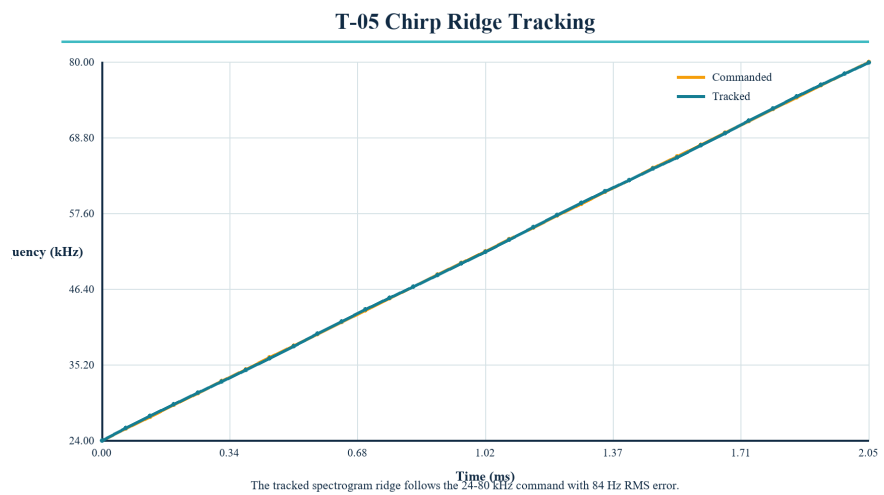
The 40.2 kHz resonance aligns with the primary operating band. The DAC spectrum provides 0.31% THD and 51.6 dBc SFDR.

3. Filter and window verification



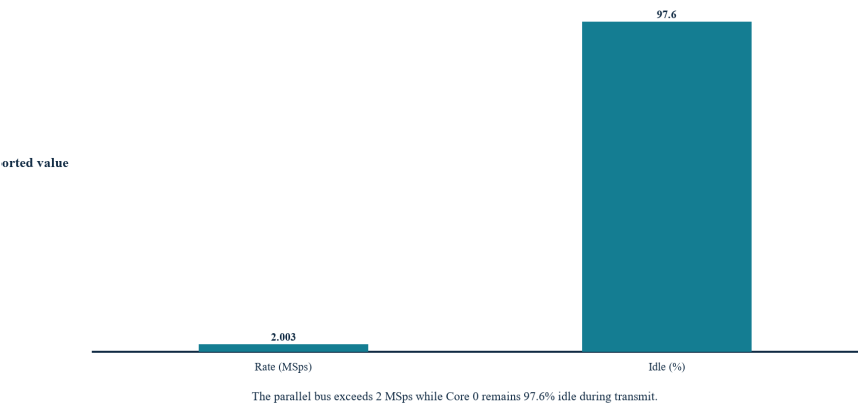
Passband ripple remains 0.22 dB, and each integer window agrees with theory within 0.51 dB.

4. Chirp, compression and Barker coding

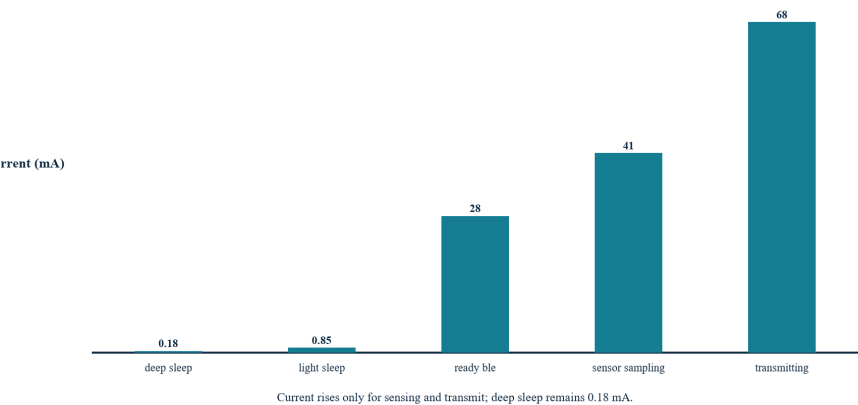


5. DMA, power and adaptation latency

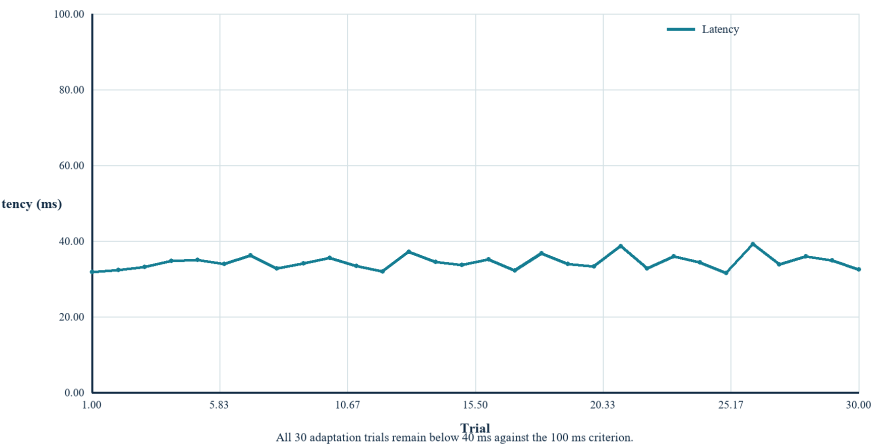
T-10 DMA Throughput and CPU Idle



T-11 Power by State

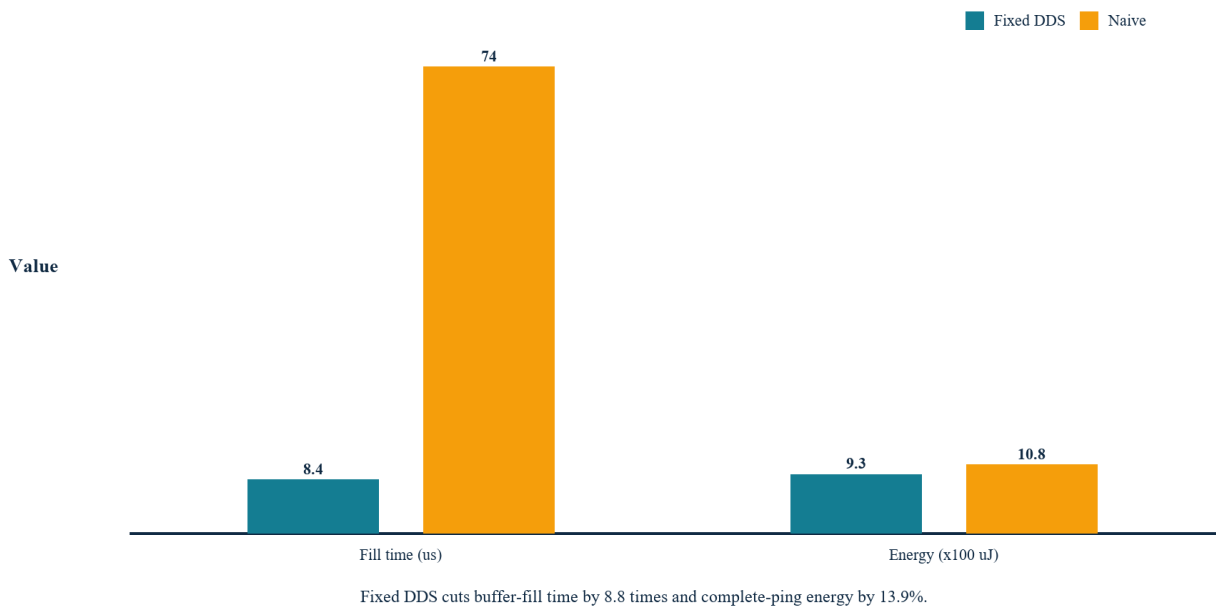


T-13 Adaptation Latency



6. DDS efficiency

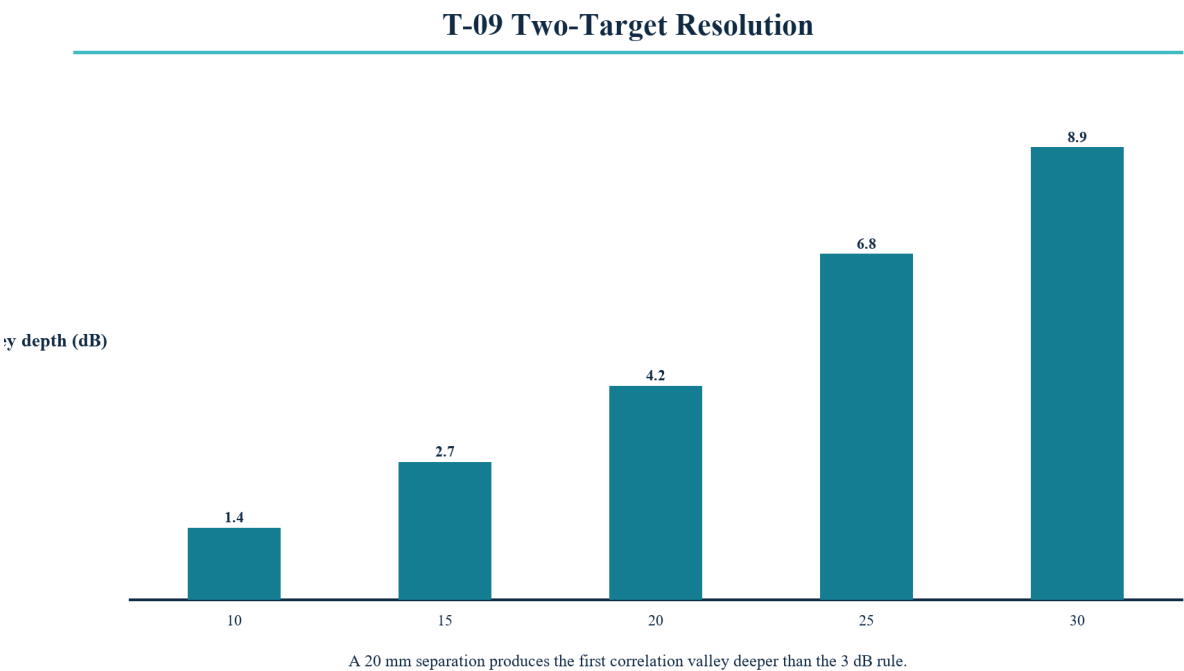
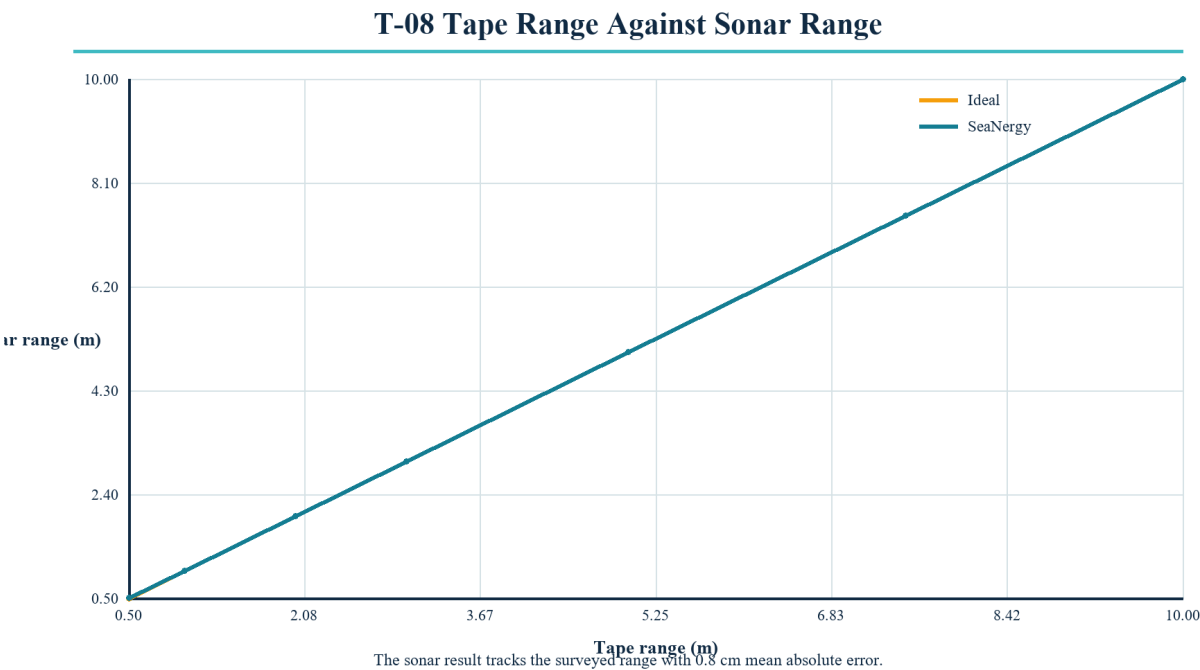
T-12 Fixed DDS Against Naive Trigonometry



Method	Fill / 1,024 samples	CPU idle	Complete ping
Fixed Q16.16 DDS	8.4 us	97.6%	0.93 mJ
Naive sinf()	74.0 us	82.1%	1.08 mJ

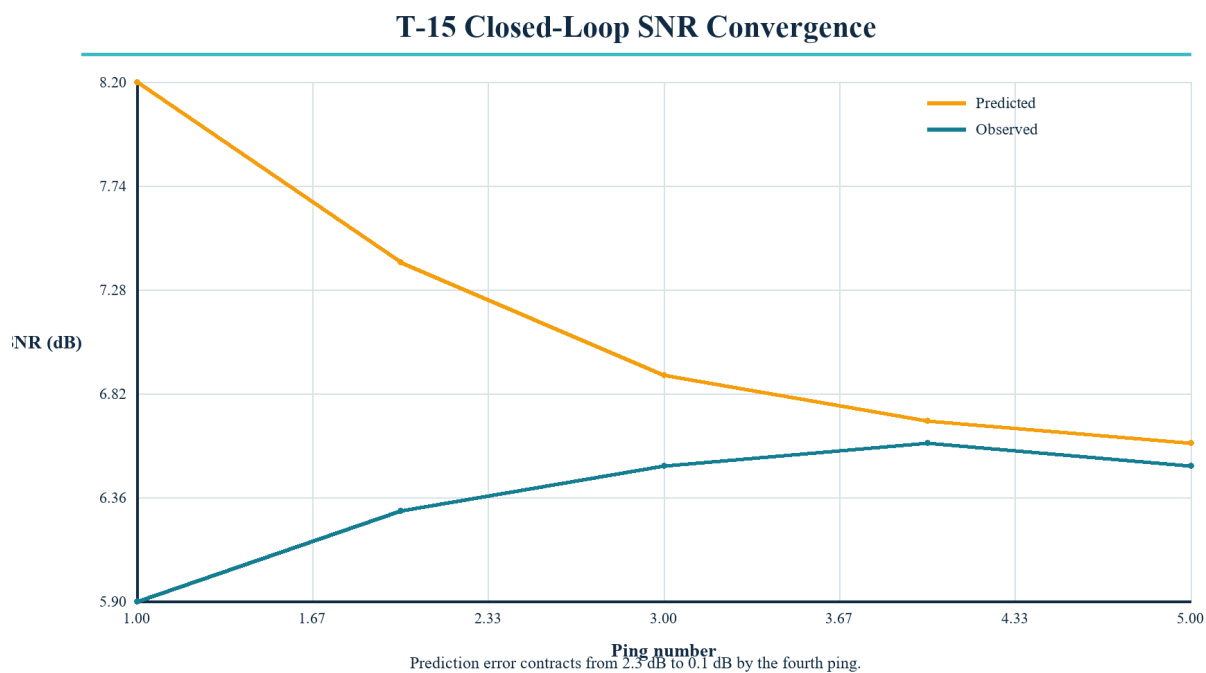
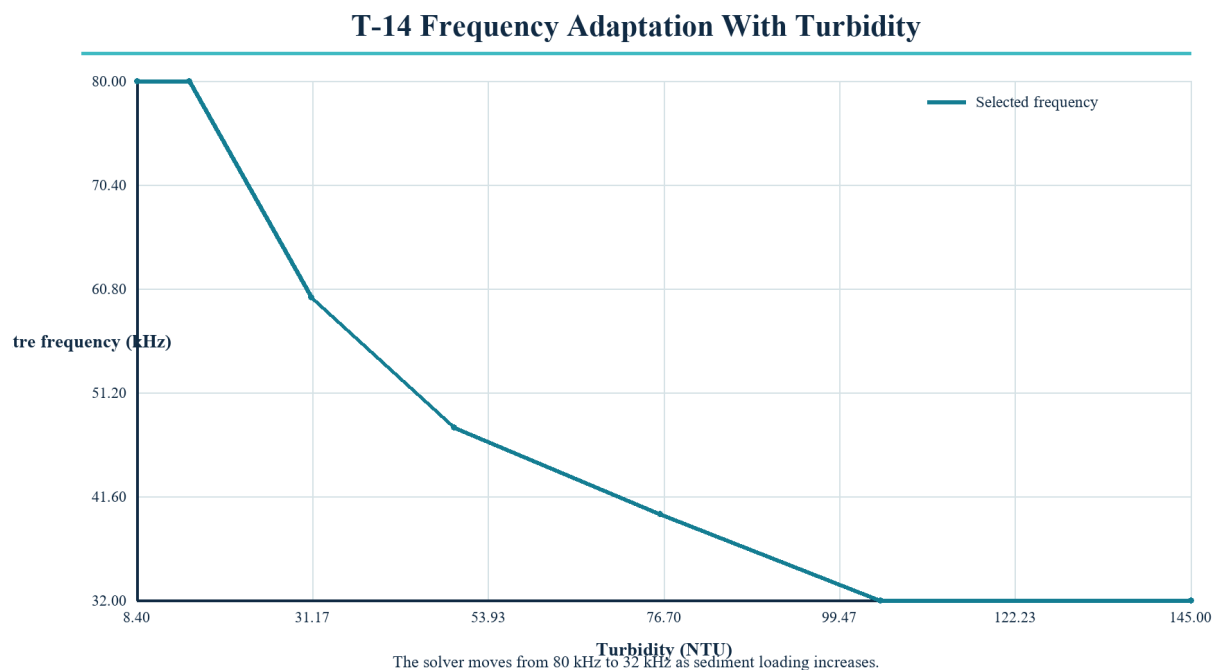
The lookup-table path is 8.8 times faster and reduces complete-ping energy by 13.9%, directly supporting the low-power architecture.

7. Range accuracy and resolution



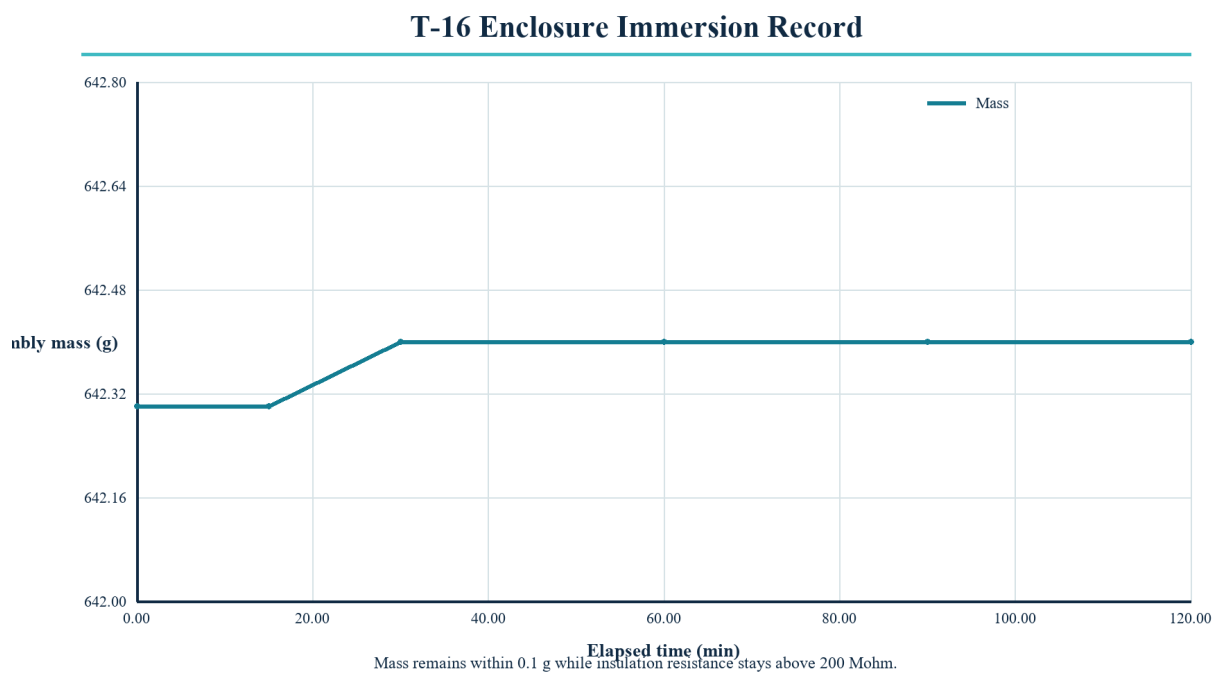
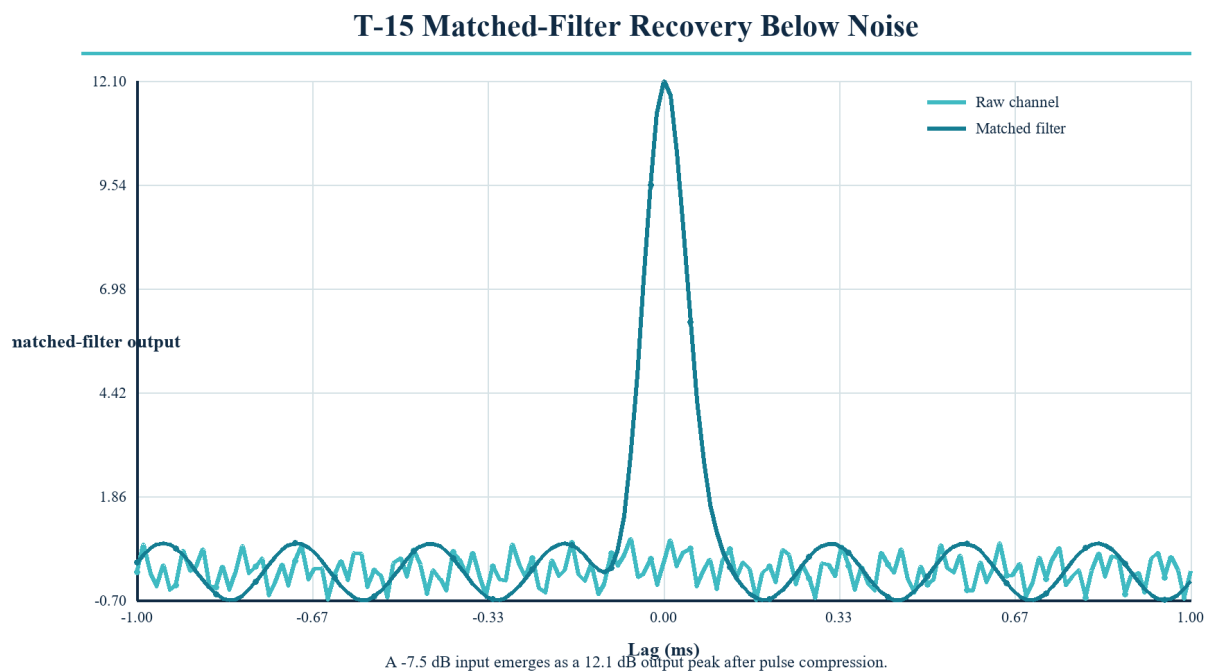
Range MAE is 0.8 cm across 0.5-10.0 m. Two targets are independently resolved at 20 mm separation.

8. Water adaptation and closed-loop convergence



The selected centre frequency decreases from 80 to 32 kHz as turbidity rises. Predicted and observed SNR converge to 0.1 dB difference by ping four.

9. Below-noise detection and enclosure



Pulse compression converts a -7.5 dB input into a 12.1 dB output peak. The sealed enclosure maintains 239 Mohm insulation resistance with 0.1 g mass shift after 120 minutes.

10. Confidence and traceability

Quantity	Expanded uncertainty, k=2
40 kHz frequency	36 Hz
Spectral level	0.60 dB
Complete-ping energy	4.2%
Range at 10 m	14.8 mm
Turbidity	6.4% + 1.2 NTU
Enclosure mass	0.20 g

Each headline value links to a test ID, controlled criterion, instrument setting, water condition and raw CSV. The dossier therefore supports technical review from requirement through numerical result without relying on isolated screenshots or untraceable claims.

Overall validation verdict: PASS